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DBMS Mini Project Synopsis

<Lead Management System>

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< 18/05/2026 >
2025-2026

For office use only:

Accepted : ☐

To be modified : ☐

Rejected : ☐

Signature of the Lab In-Charges

1. Introduction

In today's highly competitive business environment, effectively managing leads is crucial for the growth and sustainability of any organization. A Lead Management System (LMS) is a software application designed to capture, track, organize, and nurture potential customers — called leads — through the sales pipeline from first contact to conversion.

Traditional methods of managing leads using spreadsheets, paper records, or disparate tools are error-prone, time-consuming, and lack real-time visibility. A centralized, database-driven Lead Management System addresses these challenges by automating the tracking and follow-up process, assigning leads to appropriate sales agents, and generating meaningful analytics to improve decision-making.

This project implements a Lead Management System using Python as the front-end development tool and MySQL as the back-end database. The system provides role-based access for administrators, sales managers, and sales representatives, enabling efficient collaboration and lead lifecycle management.

2. Problem Statement

Organizations often struggle with managing large volumes of leads from multiple sources such as websites, social media, email campaigns, and walk-in inquiries. The key challenges include:

Lack of a centralized system to capture and store lead information from multiple sources.

Difficulty in tracking the status and history of each lead across the sales funnel.

Inability to assign and reassign leads to appropriate sales representatives efficiently.

No mechanism to set follow-up reminders or escalation alerts for overdue leads.

Absence of reporting and analytics tools to assess team performance and lead conversion rates.

Risk of data duplication, data loss, and lack of accountability without a proper database-driven system.

The proposed Lead Management System aims to solve all the above problems by providing a centralized, secure, and scalable platform that manages the complete lifecycle of a lead — from generation to closure — with proper audit trails and performance metrics.

3. Front-End and Back-End Tool Details

3.1 Front-End Tool: Python (Tkinter / Flask)

Python is used as the front-end development tool for building the user interface of the Lead Management System. Python's Tkinter library (for desktop application) or Flask framework (for web-based application) provides a clean and interactive interface for users. Key reasons for choosing Python:

Open-source, cross-platform language with extensive library support.

Easy integration with MySQL through the mysql-connector-python library.

Rapid application development with minimal code and high productivity.

Supports object-oriented programming, making the codebase maintainable and modular.

3.2 Back-End Tool: MySQL

MySQL is used as the relational database management system (RDBMS) for storing and managing all application data. Key reasons for choosing MySQL:

Open-source and widely used RDBMS with excellent performance and reliability.

Supports ACID transactions ensuring data consistency and integrity.

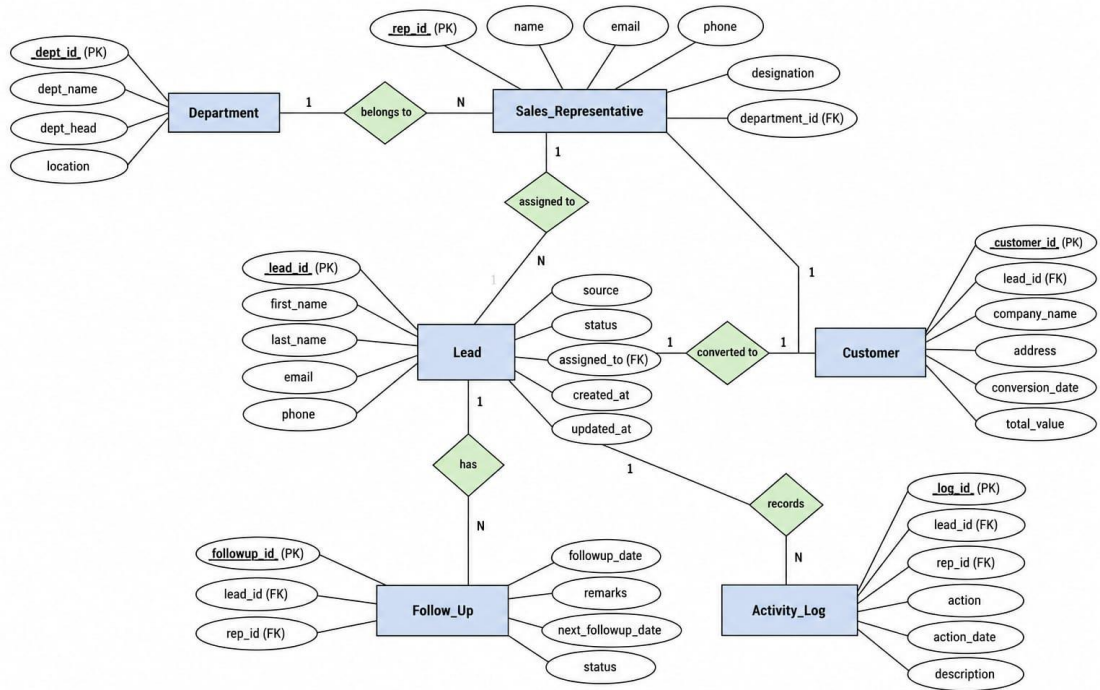
Supports stored procedures, triggers, views, and foreign key constraints.

Seamless integration with Python using mysql-connector-python or PyMySQL.

Highly scalable and suitable for both small and large datasets.

4. ER Diagram

The Entity-Relationship (ER) Diagram represents the logical structure of the Lead Management System database. The primary entities and their relationships are as follows:



Lead — Central entity with attributes: lead_id, first_name, last_name, email, phone, source, status, assigned_to, created_at, updated_at.

Customer — Linked to Lead via one-to-one relationship; attributes: customer_id, lead_id, company_name, address, conversion_date, total_value.

Sales_Representative — Manages leads; attributes: rep_id, name, email, phone, designation, dept_id.

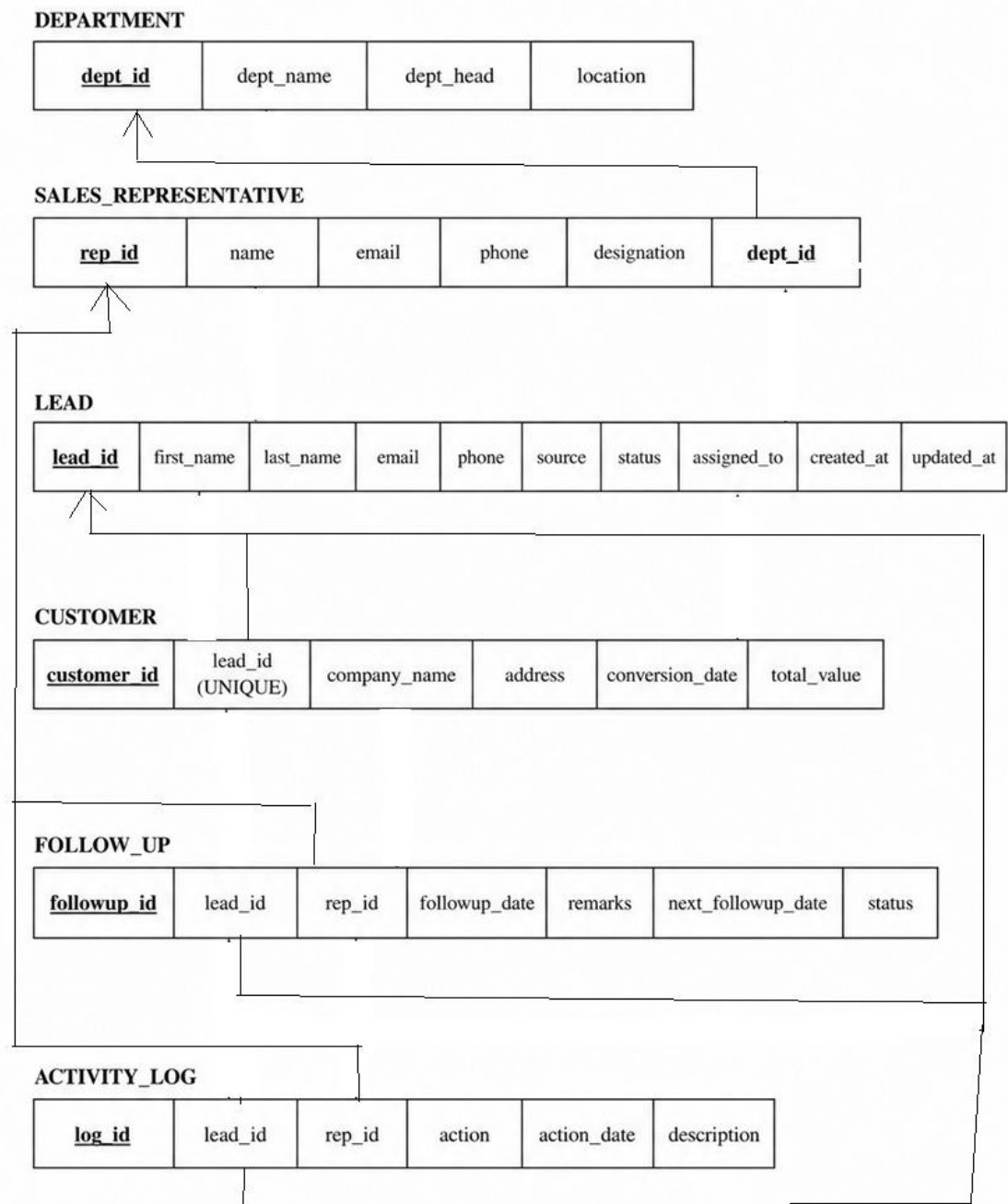
Department — Groups sales representatives; attributes: dept_id, dept_name, dept_head, location.

Follow_Up — Tracks follow-up activities per lead; attributes: followup_id, lead_id, rep_id, followup_date, remarks, next_followup_date, status.

Activity_Log — Audit log of all actions; attributes: log_id, lead_id, rep_id, action, action_date, description.

5. ER to Relational Mapping

E-R to Relational Mapping



Mapping Rules Applied:

Each strong entity is mapped to a separate table with its primary key.

Each one-to-many relationship is represented by placing the primary key of the 'one' side as a foreign key in the 'many' side table.

The one-to-one relationship between Lead and Customer is represented by placing lead_id as a foreign key in the Customer table.

The weak entity Follow_Up takes lead_id and rep_id as composite foreign keys.

6. Modules

The Lead Management System is divided into the following functional modules, each developed using Python as the front-end tool with MySQL as the database:

6.1 User Authentication Module

Login and logout functionality with role-based access control.

Roles: Administrator, Sales Manager, Sales Representative.

Password hashing and session management for security.

6.2 Lead Capture and Management Module

Add new leads with details like name, contact information, source, and initial status.

View, search, filter, and sort leads by various parameters (status, date, source, assigned rep).

Edit and update lead information as the lead progresses through the pipeline.

Delete or archive closed/lost leads.

6.3 Lead Assignment Module

Assign leads to specific sales representatives from the admin or manager panel.

Reassign leads when a representative is unavailable or overloaded.

View workload distribution across sales team members.

6.4 Follow-Up and Activity Tracking Module

Schedule follow-up calls, meetings, or emails with date and remarks.

Set reminders for upcoming follow-ups and alerts for overdue ones.

Log all activities performed on a lead including calls, emails, and meetings.

6.5 Conversion and Customer Management Module

Mark qualified leads as converted customers.

Store customer details including company name, address, and deal value.

Track the total revenue generated from converted leads.

6.6 Reporting and Analytics Module

Generate reports on lead source analysis, conversion rates, and sales rep performance.

View dashboards with key metrics like total leads, open leads, converted leads, and revenue.

Export reports to PDF or CSV format for management review.

7. Applications

The Lead Management System has a wide range of real-world applications across various industries:

Sales and Marketing Teams: Streamlines the lead-to-customer conversion process for B2B and B2C businesses.

Real Estate: Tracks property inquiries from prospective buyers and assigns agents for follow-ups.

Banking and Finance: Manages loan inquiries, insurance leads, and investment interest from potential clients.

Healthcare: Tracks patient inquiry leads for hospitals, clinics, and health service providers.

Education: Manages student enrollment inquiries and follow-up for admissions teams.

IT and Software Companies: Tracks product demo requests, free trial sign-ups, and enterprise sales inquiries.

E-Commerce: Captures leads from marketing campaigns and tracks buyer interest for upselling.

8. Back End Design

The back-end of the Lead Management System is designed using MySQL. Below are the key components of the back-end design:

8.1 Tables (Relations)

The database consists of the following core tables:

leads — Stores all lead records with status, source, and assignment details.

customers — Stores converted customer data linked to the leads table.

sales_representatives — Stores details of sales team members.

departments — Stores department information for each sales representative.

follow_ups — Stores all follow-up records for each lead.

activity_log — Stores all activity events performed on leads.

8.2 Stored Procedure

Stored Procedure: GetLeadSummaryByRep — Retrieves a summary of leads assigned to a specific sales representative, including total leads, open leads, and converted leads.

```
DELIMITER //
CREATE PROCEDURE GetLeadSummaryByRep(IN repID INT)
BEGIN
    SELECT rep_id, COUNT(*) AS total_leads,
           SUM(status='Open') AS open_leads,
           SUM(status='Converted') AS converted_leads
    FROM leads WHERE assigned_to = repID;
END //
```

8.3 Trigger

Trigger: after_lead_insert — Automatically inserts an entry into the activity_log table whenever a new lead is added to the leads table, ensuring complete audit trails.

```
CREATE TRIGGER after_lead_insert
AFTER INSERT ON leads FOR EACH ROW
BEGIN
    INSERT INTO activity_log(lead_id, rep_id, action,
                             action_date, description)
    VALUES (NEW.lead_id, NEW.assigned_to, 'Lead Created', NOW(),
            'New lead added.');
```

END;

9. Conclusion

The Lead Management System is a robust, database-driven application that addresses the core challenges faced by sales teams in managing leads effectively. By integrating Python as the front-end and MySQL as the back-end, the system provides a seamless, user-friendly interface with powerful data management capabilities.

The system ensures data integrity through proper use of primary keys, foreign keys, stored procedures, and triggers. With features spanning lead capture, assignment, follow-up tracking, conversion management, and reporting, the Lead Management System is a comprehensive solution suited for organizations across multiple industry verticals.